

10-Machine 39-Bus Power System

1 General Description

The 10-machine 39-bus power system consists of 39 buses, 10 generators, 19 loads, 34 lines and 12 transformers. A one-line diagram of the 10-machine 39-bus power system is provided in Fig. 1.

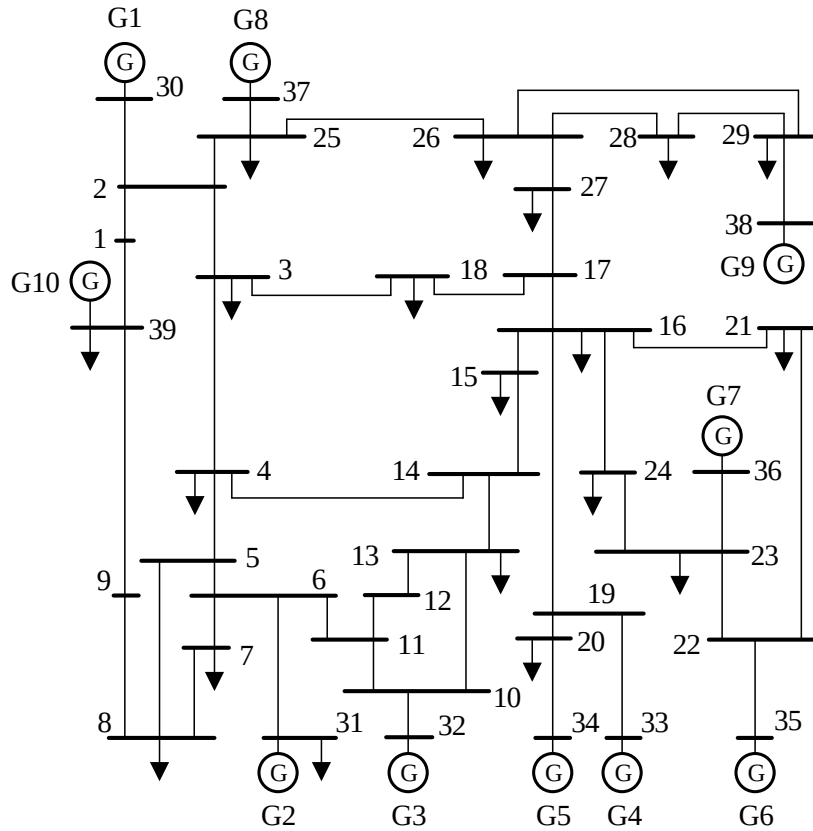


Fig. 1. The 10-machine 39-bus power system

2 Simulation Data

2.1 Data of Loads

TABLE I

Data of Loads

Load	P in MW	Q in Mvar
Load 03	322.0	2.4
Load 04	500.0	184.0
Load 07	233.8	84.0
Load 08	522.0	176.0
Load 12	7.5	88.0
Load 15	320.0	153.0
Load 16	329.0	32.3
Load 18	158.0	30.0
Load 20	628.0	103.0
Load 21	274.0	115.0
Load 23	247.5	84.6
Load 24	308.6	-92.2

Load 25	224.0	47.2
Load 26	139.0	17.0
Load 27	281.0	75.5
Load 28	206.0	27.6
Load 29	283.5	26.9
Load 31	9.2	4.6
Load 39	1104.0	250.0

2.2 Data of Lines

TABLE II
Data of Lines

From bus	To bus	R in p.u.	X in p.u.	B in p.u.
1	2	0.0035	0.0411	0.6987
1	39	0.0010	0.0250	0.7500
2	3	0.0013	0.0151	0.2572
2	25	0.0070	0.0086	0.1460
2	30	0.0000	0.0181	0.0000
3	4	0.0013	0.0213	0.2214
3	18	0.0011	0.0133	0.2138
4	5	0.0008	0.0128	0.1342
4	14	0.0008	0.0129	0.1382
5	6	0.0002	0.0026	0.0434
5	8	0.0008	0.0112	0.1476
6	7	0.0006	0.0092	0.1130
6	11	0.0007	0.0082	0.1389
6	31	0.0000	0.0250	0.0000
7	8	0.0004	0.0046	0.0780
8	9	0.0023	0.0363	0.3804
9	39	0.0010	0.0250	1.2000
10	11	0.0004	0.0043	0.0729
10	13	0.0004	0.0043	0.0729
10	32	0.0000	0.0200	0.0000
12	11	0.0016	0.0435	0.0000
12	13	0.0016	0.0435	0.0000
13	14	0.0009	0.0101	0.1723
14	15	0.0018	0.0217	0.3660
15	16	0.0009	0.0094	0.1710
16	17	0.0007	0.0089	0.1342
16	19	0.0016	0.0195	0.3040
16	21	0.0008	0.0135	0.2548
16	24	0.0003	0.0059	0.0680
17	18	0.0007	0.0082	0.1319
17	27	0.0013	0.0173	0.3216
19	20	0.0007	0.0138	0.0000
19	33	0.0007	0.0142	0.0000
20	34	0.0009	0.0180	0.0000
21	22	0.0008	0.0140	0.2565
22	23	0.0006	0.0096	0.1846
22	35	0.0000	0.0143	0.0000
23	24	0.0022	0.0350	0.3610
23	36	0.0005	0.0272	0.0000
25	26	0.0032	0.0323	0.5310
25	37	0.0006	0.0232	0.0000
26	27	0.0014	0.0147	0.2396

26	28	0.0043	0.0474	0.7802
26	29	0.0057	0.0625	1.0290
28	29	0.0014	0.0151	0.2490
29	38	0.0008	0.0156	0.0000

2.3 Data of Synchronous Generators

TABLE III

Data of synchronous generators based on 100MVA

Unit No.	H in s	D/H	x_d in p.u.	x'_d in p.u.	T'_{d0} in s	K_A	T_A in s
1	30.3	0.2	0.2950	0.0697	6.560	5	0.02
2	35.8	0.2	0.2495	0.0531	5.700	5	0.02
3	28.6	0.2	0.2620	0.0436	5.690	5	0.02
4	26.0	0.2	0.6700	0.1320	5.400	5	0.02
5	34.8	0.2	0.2540	0.0500	7.300	5	0.02
6	26.4	0.2	0.2950	0.0490	5.660	5	0.02
7	24.3	0.2	0.2900	0.0570	6.700	5	0.02
8	34.5	0.2	0.2106	0.0570	4.790	5	0.02
9	42.0	0.2	0.1000	0.0310	10.200	5	0.02
10	500.0	0.2	0.0200	0.0060	7.000	5	0.02

2.4 Data of GFM converter

TABLE IV

Data of GFM converter based on 100MVA

Parameters	Values
U_b in V	575
M in s	20
K_i in s	0.1
K_q	5
T_u in s	0.02
K_u	1

3 Model of Generator

3.1 Model of Synchronous Generator

The model of synchronous generator is

$$\begin{cases} \dot{\delta} = \omega_0 \omega \\ 2H \dot{\omega} = P_M - P_e - D\omega \\ T'_{d0} \dot{E}'_q = E_{fd} - E'_q - (x_d - x'_d)I_d \\ T_A \dot{E}_{fd} = K_A(V_{ref} - |V_t|) - E_{fd} \end{cases}$$

3.2 Model of GFM Converter

The model of GFM converter is

$$\begin{cases} \dot{\delta} = \omega_0 \omega \\ M \dot{\omega} = P_M - P_e - D \omega \\ K_i \dot{E}_{vir} = E_{vir_fd} + K_q (Q_{ref} - Q_e) \\ T_u \dot{E}_{vir_fd} = K_u (V_{ref} - |V_t|) - E_{vir_fd} \end{cases}$$